viewing tool, PCB-Investigator is totally scriptable and customisable through its automation interface. It can be used to view ODB++, GenCAD, IPC-2581, Gerber, Excellon, Sieb & Meyer and DXF formats in addition to images. This makes it easier for the user to export data from all popular CAD design tools such as Mentor Graphics, OrCAD and Altium Designer.

Using PCB-Investigator, users can enhance the hardware for PCB manufacture by visually finding nets, components, test points and connectors; perform EMC analysis; and optimise assembly for PCB manufacture. Other essential features include heat analysis by finding hotspots and bus optimisation. Users can change tracking by comparing different indices and variants of a printed circuit board, bringing together all developing departments by simply passing information.

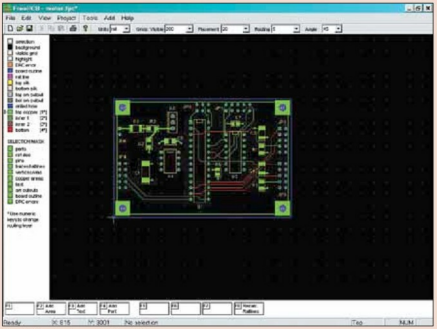


Fig. 4: Creating layout in FreePCB

Liquid PCB

Liquid PCB is a computer-aided design application for PCBs. It is written in C++, and uses wxWidgets and OpenGL. This software is ideal for users who design a lot of PCBs with strict space constraints and awkward shapes.

While giving users the freedom to produce an excellent design, Liquid PCB makes it possible to navigate the interface using as few mouse clicks as possible. It actually uses the immense power of a computer's CPU and GPU to help users. Liquid PCB constantly enforces design rules, and optimises track, component and via placement on the PCB.

The software allows users to drag tracks wherever they want and even squeeze them into small gaps. Its user interface has been designed specifically to reduce the amount of clicking and fiddling required by users. Many functions can be accessed with only one click, making it easier to navigate and design a PCB.

FreePCB

FreePCB is a user-friendly intuitive software for PCB design. It supports a slightly different file format from other conventional PCB design software, so users migrating from other such software might feel the need to adjust a little. Other than that, it basically shares the features of other PCB software.

The software supports up to 16 copper layers, both metric and English units, and export of designs in RS-274X Gerber format. Autorouting is possible both partially or fully by using the FpcROUTE Specctra DSN file translator.

FreePCB accepts a netlist from any schematic editing program that can output a PADS format netlist. The user can then manually place the components in FreePCB and route traces between pins of the components. FreePCB also interfaces to FreeRoute auto-router program, relieving the user from having to manually route each trace in the design. It outputs RS-274X formatted Gerber file data and a drill file. If desired, a schematic is not necessary and the user can create a design with FreePCB from scratch by adding components and nets manually.

PCBColorizer

PCBColorizer software is designed to increase the accessibility of components on a PCB layout. It creates automatic colouring schemes for a PCB, wherein a separate colour is allocated to each type of component. This makes it simple to visually search components on the board. It also generates a text list of components that shows which type of component is associated with what colour.

PCB

PCB is an interactive printed circuit board editor capable of running on Windows, Linux and Mac systems. With a fully integrated solution for PCB design, it offers users an alternative to proprietary PCB design software. PCB includes a rats nest feature, schematic/netlist import, design rule checking, and can provide industry-standard RS-274X (Gerber), NC drill, and centroid data (X-Y data) output for use in the board fabrication and assembly process, as well as photo-realistic and design review images.

PCB features an autorouter and trace optimiser, which can reduce layout time tremendously. For custom requirements, it offers a plug-in application programming interface for inserting new functionality and using that functionality from within the GUI as well as in scripts. The software offers both GUI and command line options, providing more operational freedom to users.

GerberLogix

GerberLogix is a PCB inspection tool for electronics professionals. It supports a number of file formats includ-